Project Proposal: Graduate Independent Study in Computer Science

Project Title: Vector Database Comparison for Semantic Search Performance
Student Name: Shivali Mate
Faculty Mentor: Professor Feng Chen
Course: Graduate Independent Study (Computer Science)
Credit Hours: 3
Semester: Spring 2026

1. Work Plan

This independent study will explore methods for evaluating and comparing vector database systems in performing semantic search and information retrieval. The project will involve processing diverse datasets, transforming content into vector representations to embed them into the vector databases, and designing workflows to manage data storage and querying which will be assessed for various performance metrics. Additional components, such as language models or RAG systems, may be incorporated to examine further capabilities and assess the impact on retrieval or question answering. The expected outcome is a functional evaluation framework supported by clear documentation and performance evaluation metrics.

2. Expected Student Work Hours

The student will dedicate 9-10 hours per week to the study. This time includes reading current literature, database setup and experimentation, coding, documentation, and meetings with the faculty mentor. This aligns with the standards for a 3 credit graduate independent study.

3. Required Meetings with Mentor

- Regular Weekly Meeting: One meeting per week with the faculty mentor, scheduled for progress updates, troubleshooting, and milestone reviews.
- Ad-hoc consultations as needed for technical or scope adjustments.

4. Required Readings and Assignments

As a research based study, assignments will be milestone driven rather than a fixed set of tasks.

Readings:

Core readings will be selected and reviewed by the student, focusing on vector databases, semantic search, and benchmarking methodology. Literature review will be ongoing, drawing from academic and industry sources relevant to modern database research.

Assignments & Milestones:

The project will be broken down into phases. Specific tasks and deadlines will be determined collaboratively with the mentor and reviewed during weekly meetings.

5. Assessment and Grading Criteria

The final grade will be determined based on the student's consistent engagement, the quality of their contributions, and their ability to document and communicate their work effectively.

- **Weekly Progress and Participation:** Assessed based on the student's clear and prepared updates during weekly group meetings, active participation in discussions, and consistent progress toward agreed upon milestones.
- **Technical Contribution & Execution:** Evaluated on the quality, functionality, and rigor of the code, models, and analyses produced. This includes the successful implementation of assigned tasks.
- **Documentation and Final Reporting:** Graded on the clarity, completeness, and quality of technical documentation, including code comments, slides for presentations, and a final summary report of the project work and results.